

Station 1

In human anatomy, the mouth is the first portion of the alimentary canal that receives food and produces saliva. The oral mucosa is the mucous membrane epithelium lining the inside of the mouth. The mouth also helps in chewing and biting food and contains the strongest muscle in the body for its size and only muscle in the body attached only on one end (the tongue).

The Mouth: A Health Context



Oral cancer, also known as mouth cancer, is a type of head and neck cancer and is any cancerous tissue growth located in the oral cavity. It may arise as a primary lesion originating in any of the tissues in the mouth, by metastasis from a distant site of origin, or by extension from a neighboring anatomic structure, such as the nasal cavity. Alternatively, the oral cancers may originate in any of the tissues of the mouth, and may be of varied histologic types. Around 75 percent of oral cancers are

linked to modifiable behaviors such as tobacco use and excessive alcohol consumption. Other factors include poor oral hygiene, irritation caused by ill-fitting dentures and other rough surfaces on the teeth, poor nutrition, and some chronic infections caused by fungi, bacteria or viruses. In 2011, close to 37,000 Americans are projected to be diagnosed with oral or pharyngeal cancer. 66% of the time, these will be found as late stage three and four disease. It will cause over 8,000 deaths. Of those newly diagnosed, only slightly more than half will be alive in five years.

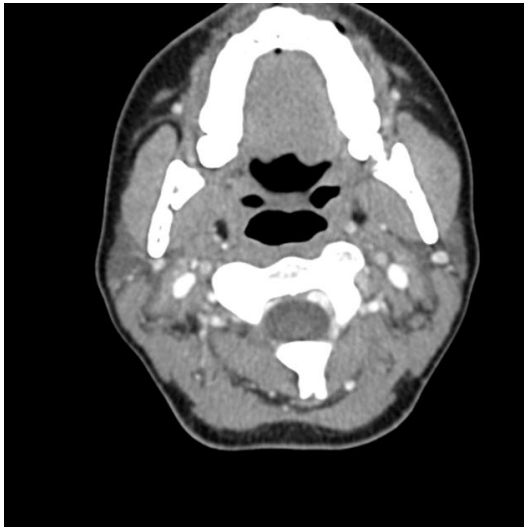
Many different cell types in the mouth can become cancerous. Below are examples of a few common cell types in oral cancer:

Squamous cell carcinoma: More than 90 percent of cancers that occur in the oral cavity and oropharynx are squamous cell carcinoma. Normally, the throat and mouth are lined with so-called squamous cells, which are flat and arranged in a scale-like way. Squamous cell carcinoma means that some squamous cells are abnormal.

Minor salivary gland carcinomas: This category includes several kinds of oral cancer that can develop on the minor salivary glands, which are found throughout the lining of the mouth and throat. These types include adenoid cystic carcinoma, mucoepidermoid carcinoma, and polymorphous low-grade adenocarcinoma.

Lymphomas: Oral cancers that develop in lymph tissue, which is part of the immune system, are known as lymphomas. The tonsils and base of the tongue both contain lymphoid tissue. See our pages on Hodgkin lymphoma and non-Hodgkin lymphoma for cancer information related to lymphomas in the oral cavity.

Normal



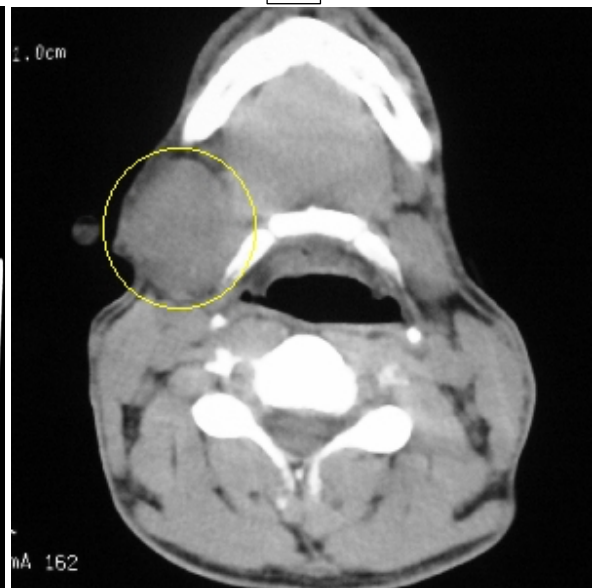
1



2



3



Station 2

The human teeth function to mechanically break down items of food by cutting and crushing them in preparation for swallowing and digesting. Humans have four types of teeth: incisors, canines, premolars, and molars, each with a specific function. The incisors cut the food, the canines tear the food and the molars and premolars crush the food. The roots of teeth are embedded in the maxilla (upper jaw) or the mandible (lower jaw) and are covered by gums.

Humans, like other mammals, develop two sets of teeth. The first set (called the "baby", "milk", "primary", or "deciduous" set) normally starts to appear at about six months of age, although some babies are born with one or more visible teeth, known as natal teeth. Normal tooth eruption at about six months is known as teething and can be painful.

Enamel is the outer most material of a tooth, and is the hardest and most highly mineralized substance of the body. Dentin is the substance between enamel and the pulp chamber. The porous, yellow-hued material is made up of 70% inorganic materials, 20% organic materials, and 10% water by weight. Because it is softer than enamel, it decays more rapidly and is subject to severe cavities if not properly treated. The dental pulp is the central part of the tooth filled with soft connective tissue. This tissue contains blood vessels and nerves that enter the tooth from a hole at the apex of the root.

Teeth: A Health Context

Tooth decay, also known as dental caries or cavities, is a breakdown of teeth due to acids made by bacteria. The cavities may be a number of different colors from yellow to black. Symptoms may include pain and difficulty with eating. Complications may include inflammation of the tissue around the tooth, tooth loss, and infection or abscess formation. The following page includes x-rays (or pictures) of a number of cavities from different individuals. Cavities generally appear as darker colored areas on a tooth due to missing enamel (which is light colored) and are indicated by arrows or circles.

*Normal Tooth X-Ray, Showing
How Layers Appear:*

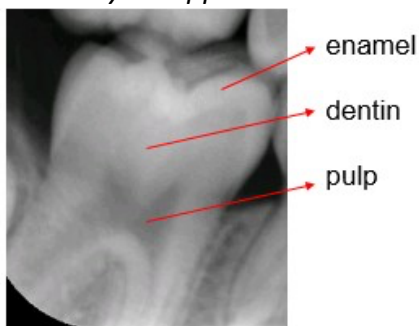


Image 1

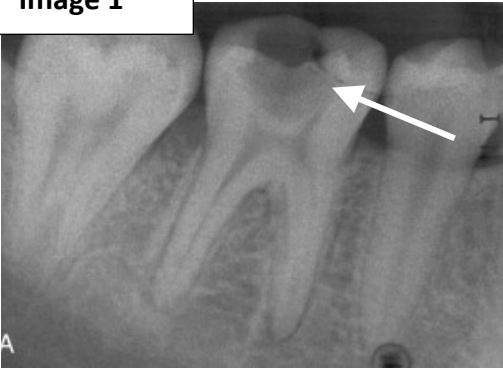


Image 2

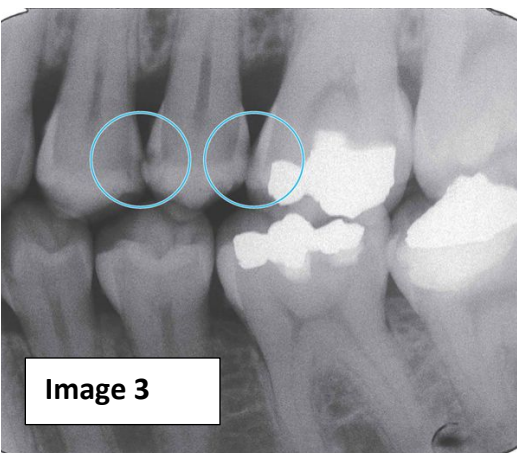


Image 3



Image 4

Image 5

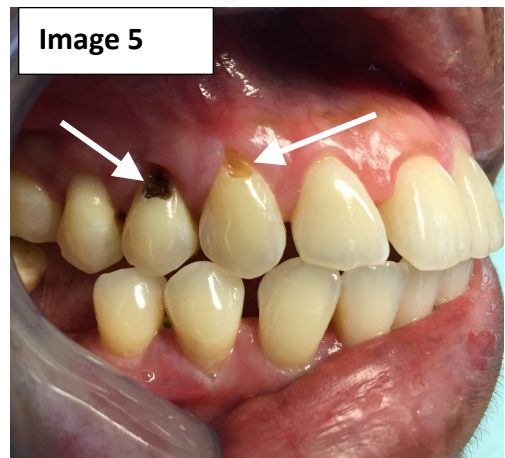


Image 6



Station 3

The esophagus, commonly known as the food pipe or gullet (gut), is an organ through which food passes, aided by peristaltic contractions, from the pharynx to the stomach. The esophagus is a fibromuscular tube, about 25 centimetres long in adults, which travels behind the trachea and heart, runs parallel to the descending aorta, passes through the diaphragm and empties into the uppermost region of the stomach. During swallowing, the epiglottis tilts backwards to prevent food from going down the larynx and lungs. The lower esophageal sphincter, or gastroesophageal sphincter, surrounds the lower part of the esophagus at the junction between the esophagus and the stomach. Dysfunction of the gastroesophageal sphincter causes gastroesophageal reflux, which causes heartburn and if it happens often enough, can lead to gastroesophageal reflux disease, with damage of the esophageal mucosa

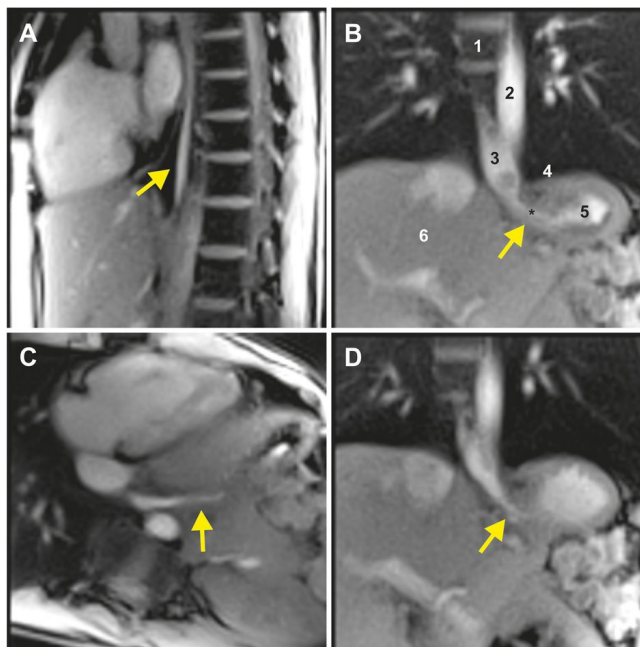
The Esophagus: A Health Connection



Gastroesophageal reflux disease (GERD), also known as acid reflux, is a long term condition in which stomach contents (white material in image to left) rise up into the esophagus, resulting in either symptoms or complications. Symptoms include the taste of acid in the back of the mouth, heartburn, bad breath, chest pain, vomiting, breathing problems, and wearing away of the teeth. Complications include esophagitis, esophageal stricture, and Barrett's esophagus. GERD is caused by a failure of the lower esophageal sphincter. In healthy patients, a valve prevents duodenal bile, enzymes, and stomach acid from traveling back into the esophagus where they can cause burning and inflammation of sensitive esophageal tissue. The burning and pain is sensed by pain receptors in the

esophagus wall.

In the most superior part of esophagus, pain receptors are abundant in the submucosa and mucosa and some fibers terminate in the epithelium. However, the innervation becomes progressively sparse in the thoracic and abdominal esophagus. The most dense regions of pain receptors are consistently found between the muscularis externa and the muscular mucosae along the whole length of the esophagus. (source: <https://www.nature.com/gimo/contents/pt1/full/gimo16.html>)



Station 4

The stomach is a muscular, hollow organ located between the esophagus and the small intestine. It secretes digestive enzymes and gastric acid to aid in digestion. The pyloric sphincter controls the passage of partially digested food (chyme) from the stomach into the duodenum where peristalsis takes over to move this through the rest of the intestines. Unlike other parts of the digestive tract, the stomach has an extra, oblique layer of muscle in the muscularis externa. This provides the stomach with extra churning power to pummel food into smaller pieces.

The Stomach: A Health Connection



A gastric ulcer is a break in the lining of the stomach. Symptoms include waking at night with upper abdominal pain or upper abdominal pain that improves (or sometimes worsens) with eating. The pain is often described as a burning or dull ache. Other symptoms include belching, vomiting, weight loss, or poor appetite. Common causes include certain bacterial infections and use of NSAIDs (drugs like ibuprofen), which reduce mucus production and make the stomach more sensitive to its acids. If severe and untreated, ulcers can lead to perforations, or holes in the wall of the stomach (shown to left). This can lead to serious and sometimes fatal

consequences due to bleeding and leakage of stomach contents into the abdomen.

Station 5

The large intestine, also known as the large bowel, is the last part of the gastrointestinal tract and of the digestive system. Water is absorbed here and the remaining waste material is stored as feces before being removed by defecation. The colon is the largest portion of the large intestine, so many mentions of the large intestine and colon overlap in meaning whenever anatomic precision is not the focus. Most sources define the large intestine as the combination of the cecum, colon, rectum, and anal canal. Of note: In cases of cancer or severe inflammatory bowel disease, the large intestine can be surgically removed without any major impacts on digestive activity.

The Large Intestine: A Health Context

Colorectal cancer is the second highest cause of cancer occurrence and death for men and women in the United States combined. Colorectal tumors often start as non-cancerous polyps (picture to right). As a result, a preventative measure for colorectal cancer is to screen for polyps through colonoscopy.

Colonoscopy is the endoscopic examination of the large intestine and the distal part of the small bowel with a fiber optic camera on a flexible tube passed through the anus. It can provide a visual diagnosis (e.g. ulceration, polyps) and grants the opportunity for biopsy or removal of suspected colorectal cancer lesions.

Colonoscopy can remove polyps as small as one millimetre or less. Once polyps are removed, they can be studied with the aid of a microscope to determine if they are precancerous or not.

